

"PAPER_{0:D3}" -f [int]# PAPER #74 □ Galactic Structure: NED + SIMBAD + UQFF Cross-Validation

Title: Galactic Structure Cross-Validation: NED and SIMBAD Multi-Object Queries vs UQFF Predicted Velocity Dispersions and Mass Profiles

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ($\gamma = 0.0005/\text{day}$, $[\text{SSq}] = 0.57$)

Date: March 7, 2026

Source Data: QCalc_validation.py (DataSourceURLs: NED_BASE, SIMBAD_BASE, SIMBAD_API)

Index Slot: §1.10 Database Integration & Multi-Wavelength Astrophysics, \$n = [int]# PAPER #74 □ Galactic Structure: NED + SIMBAD + UQFF Cross-Validation

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NED TAP ADQL Query (Galaxy Velocity Dispersions)

From QCalc_validation.py

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$$\sigma_{\text{UQFF}}^2 = \sigma_{\text{Newton}}^2 \times \left(1 + \frac{F_{U,Bi,i}}{F_{\text{Newton}}} \right) = \frac{GM}{r_{\text{eff}}} \times \left(1 + \frac{\Omega_g \cdot \sum U g_j}{GM/r^2} \right)$$

Validation Results by Galaxy Type

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Average UQFF enhancement: $s_UQFF/s_Newton = \mathbf{1.018}$ ($= [SSq] \times 0.032$ correction factor).

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SIMBAD provides stellar/galactic spectral types and radial velocities. The UQFF predicts no modification to radial velocities (cosmological redshift is Hubble-standard) but does predict a 0.57% perturbation to measured proper motions in galaxies with active AGN core fields:

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For M31 (Andromeda): $r_{AGN}/r_{gal} \sim 0.001$, so $d\mu \sim 0.057\%$ within SIMBAD proper motion uncertainties ($> 10\%$) for extragalactic objects.

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When SIMBAD + NED + GAIA data are combined for the same galaxy:

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Proper motion	$\square$	$\square$	?	$+d\mu$ (negligible)

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Database	Query Method	UQFF Prediction	Agreement
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Database	Query Method	UQFF Prediction	Agreement
Combined	Cross-match	Consistent systematic +1.8%	Self-consistent

Source: *QCalc_validation.py* *NED_BASE* + *SIMBAD_BASE* endpoints | ? = 0.0005/day
| [SSq] = 0.57 .Groups[1].Value "PAPER_{0:D3}" -f \$n

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Source: `QCalc_validation.py` `NED_BASE + SIMBAD_BASE endpoints` | $? = 0.0005/\text{day}$ | $[SSq] = 0.57$.Groups[1].Value $\square$ Galactic Structure: NED + SIMBAD + UQFF Cross-Validation

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Index Slot: §1.10 Database Integration & Multi-Wavelength Astrophysics, $\$n = [\text{int}] \# \text{“PAPER_}\{0:\text{D}3\}” -f [\text{int}] \# \text{PAPER \#74}$ $\square$ Galactic Structure: NED + SIMBAD + UQFF Cross-Validation

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UQFF computed: Galactic scale UQFF gravity correction $g_{\text{UQFF}}/g_{\text{Newton}} = 1 + [SSq] \times ? \times (r/\text{kpc})^2 = 1 + 2.85\text{e-}4 \times (8.5)^2 = 1.0206\text{e+}0$; 2.06% deviation at Galactic Center.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by `upgrade_early_whitepapers.py` (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
$\hat{\Gamma}^o$	$5.0 \times 10^{-6} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
$\hat{\Gamma}^2_i$	0.60–0.61	Buoyancy coupling coefficient
k_{11}	1.5	Ug1 DPM-dipole coupling
k_{12}	1.2	Ug2 outer-bubble charge coupling
k_{13}	1.8	Ug3 string-rotation coupling
k_{14}	2.0	Ug4 vacuum-concentration coupling
$\hat{\mathbf{I}} \cdot$	10^{-2} Å^2	Inertia tensor scale
E_react(0)	10^{-6} eV	Reference reactive energy

A.2 F U Master Equation (Complete $\hat{\rho}$ 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 i g l [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}} i g r]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heavyside-amplified)	compute_UM_SOURCE4 / compute_UM()
âĀĤÎĬ»âĀĢÂ-ŰâĤÏĦ	4th Dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER 420):

$\hat{I} \rightarrow \hat{I} + 10 \hat{a} \rightarrow \hat{A}^1 \hat{a}^0$, $\hat{I} \rightarrow \hat{I} + 10 \hat{a} \rightarrow \hat{A}^1 \hat{A}^2$, $\hat{I} \rightarrow \hat{I} + 10 \hat{a} \rightarrow \hat{A}^1 \hat{A}^1$, $\hat{I} \rightarrow \hat{I} + 10 \hat{a} \rightarrow \hat{A}^1 \hat{A}^3$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER 421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imesigl}(1 + 10^{13} \Theta(ho_{SCm} - ho_c) \text{igr}) \text{imesigl}(1 + A_g \cos(\Delta\omega t) \text{igr})$$

Symbol	Value	Description
ρ_c	$10^{14} \mu\text{g}/\text{m}^3$	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\dot{\phi}$	$2\pi/(434 \cdot 365.25) \text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	U_{g_sum} + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, $\dot{\phi}$)	Multi-scale field interactions
Buoyant	$\dot{\phi}^2_i \propto U_{bi}$	Expanding nebulae, stellar winds
Superconductive	$\propto (1 + 10^{13} \cdot f_H)$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in `MAIN_1_CoAnQi.cpp`, `CondensedPhysics.py`, and `CondensedPhysics2.py`.